MIL-G-46905A (MU)
2 MARCH 1967
SUPERSEDING
MIL-G-46905 (MU)
6 NOVEMBER 1962

MILITARY SPECIFICATION

GROMMETS, PLASTIC, FILAMENT WOUND, (FOR SEPARATE LOADED PROJECTILES)

1. SCOPE

1.1 This specification covers filament wound polyester resin impregnated, glass roving grommets for use in protecting the rotating band of separate loaded projectiles.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-I-45607 — Inspection Equipment; Supply and Maintenance of.

STANDARDS

FEDERAL

FED-STD-406 — Plastics, Method of Testing.

FED-STD-595 — Colors.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-1168 — Lot Numbering of Ammunition.

MIL-STD-1169 — Packaging, and Marking for Interplant Shipment of Inert Ammunition Components.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC 8140

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PUBLICATIONS

ARMY

ET–8863065, 8864366, 8864367, 8881587 — Equipment Tabulation.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material shall be in accordance with applicable drawings and specifications (see 6.4).

3.1.1 *Resin.* The resin bath used in impregnating the roving shall consist of; a bisphenol A derived polyester resin, (as the base resin), plus a suitable solvent, a flexible polyester resin, an accelerator, a catalyst, and a pigment (see 6 5).

3.1.2 *Grommet inserts.* The grommet inserts shall conform to all requirements specified on the applicable drawings (see 6.4).

3.2 Grommets. The grommets shall comply with all requirements specified on the applicable drawing (see 6.4) and with all requirements specified in applicable specifications.

3.3 Barcol hardness. The Barcol hardness shall be 50 minimum when tested as specified in 4.4.2.

3.4 Water absorption. The water absorption shall be 1.0 percent maximum when tested as specified in 4.4.3.

3.5 Color (see 4.3.2.3). Color of the grommet shall be within the range of color chip numbers 24064 and 24087 or gray approximating chip number 36231 of FED–STD–595 (see 6.1).

3.6 Workmanship.

3.6.1 *Metal parts.* All parts shall be dry and free of chips, burrs, sharp edges, dirt grease, and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

3 6.2 *Grommet.* The grommet shall be free from cracks, air bubbles, holes, blisters, resin starved areas, resin pockets, tackiness, and delamination or other separations.

3.7 First article inspection. This specification make provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define the terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

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(a) A statement that the lot complies with all quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 **First article inspection.**

4.2.1 *Submission.* The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 50 assemblies and 10 complete sets of parts which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized, by the Contracting Officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 *Inspection to be performed.* Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 *Rejection.* If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection

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upon any failure of an assembly or component to comply with any of the stated requirements.

4.3 Inspection provisions.

4.3.1 *Lot formation.* The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL–STD–105 and shall be numbered in accordance with MIL–STD–1168. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector.

4.3.2 *Examination.* Inspection for critical defects (and major defects, when so specified,) shall be 100 percent. Sampling plans and procedures for major and minor defects shall be in accordance with Standard MIL–STD–105 except that continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQLs and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 *Grommet insert, prior to finishing* (see applicable dwg) (see 6.4).

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2)*
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 2.5 percent		
201.	Total length	Gage	01001
202.	Height	Gage	01002
203.	Length of ears	Gage	01003
204.	Width	Gage	01004
205.	Incorrect angle	Gage	01005
206.	Evidence of poor workmanship (see 3.6.1)	Visual	01006

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4.3.2.2 *Grommet insert after finishing* (see applicable dwg) (see 6.4).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 0.65 percent		
201.	Color improper	Visual	02001

4.3.2.3 *Grommet assembly* (see applicable dwg) (see 6.4).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Grommet insert missing	Visual	03001
102.	Notch to hold grommet open missing or improper	Visual	03002
103.	Width between inside retaining edge	Gage	03003
Minor:	AQL 4.00 percent		
201.	Outside diameter	Gage	03004
202.	Smallest inside diameter	Gage	03005
203.	Angle and position of opening cuts	Gage	03006
204.	Position of inserts	Gage	03007
205.	Diameter of inside retaining edge	Gage	03008
206.	Location of ridges on inner surface (when applicable)	Gage	03009
207.	Diameter of edges on inner surface (when applicable)	Gage	03010
208.	Color (see 3.5)	Visual	03011
209.	Evidence of poor workmanship (see 3.6.2)	Visual	03012

3.4.2.4 *Container sealed.*

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.5 percent		
201.	Improper closure	Visual	04001
202.	Contents loose	Manual	04002
203.	Marking misleading or unidentifiable	Visual	04003
204.	Gross weight, maximum	Gage	04004

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4.3.3 *Testing.*

4.3.3.1 *Opening force* (see applicable dwg) (see 6.4), major defects. Ten grommets shall be selected for each test. If any specimen fails to comply with the requirement the lot shall be rejected. Code No. 05001.

4.3.3.2 *Barcol hardness* (see 3.3), major defect. Five grommets shall be selected from each lot for testing as specified in 4.4.2. Sufficient hardness impressions shall be taken to assure that the item has been properly cured. Proper curing shall be indicated if at least 5 impressions on each sample grommet evidence a Barcol hardness of 50, minimum. Failure of any sample to indicate proper curing shall be cause for rejection of the lot. Code No. 06001.

4.3.3.3 *Resin content* (applicable dwg.) (see 6.4), major defect. Five grommets shall be selected for this test. If any specimen fails to comply with the requirement the lot shall be rejected. Code No. 07001.

4.3.3.4 *Water absorption (see 3.4), major defect.* The five grommets selected in 4.3.3 3 shall be used for this test. If any specimen fails to comply with the requirement the lot shall be rejected. Code No. 08001.

4.3.4 *Inspection Equipment.* Equipment Tabulation Number ET–8863065, 8864366, 8864367 and 881587 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL–I–45607.

4.3.4.3 *Government use of contractor's inspection equipment.* The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 *Opening force.* The specimens shall be conditioned prior to testing in accordance with section 4 of FED–STD–406. The force required to open the grommet to the locked position shall be determined by means of a spring-loaded tension gage.

4.4.2 *Barcol hardness.* Each grommet shall be placed on its side. A steel block which is the same height as the grommet shall be placed in the center of the grommet. The legs of the impressor shall be supported on the steel block while the impressions are being taken. The impressions shall be taken not closer than 1 inch from any other impression, not closer than 1½ inches from either insert, and not closer than 1½ inches from the opening.

4.4.3 *Water absorption.* The water absorption shall be determined on two specimens from each of three grommets (not including the inserts) in accordance with FED–STD–406, method 7031A except the specimen shall be 2 inches long and the

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width of the grommet and only the percent weight increases shall be determined.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level C.* Unless otherwise specified by the procuring activity, all packing and marking shall be in accordance with MIL–STD–1169.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Title and number of applicable drawing.

(c) Color required.

(d) First article inspection provisions.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3 2 *Inspection equipment lists codes.* The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD — Contractor design on controlled contractor format or commercial equipment.

MU — Army design, mandatory for use.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor-prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information

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necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL–D–1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 *Submission of contractor designs.* Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical or major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND1. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 **Drawings.** The following is a list of the applicable drawings at the time of publication:

8863065 — Grommet, for Shell, 155MM, Type I.

8864366 — Grommet, for Shell, 175MM, Type I.

8864367 — Grommet, for Shell, 8 inch, Type I.

8881587 — Grommet, for Shell, 155MM, Type II.

6.5 The following resin formulation has been found suitable for use as the impregnating bath (all percentages are by weight).

Material	*Percent*
Atlac 382 (or equivalent)	38.2
Styrene monomer	48.0
Plaskon 9600 (or equivalent)	9.6
Nuodex (6 percent cobalt naphthenate)	0.4
Pigment	2.8
Methyl ethyl ketone peroxide	1.0

6.6 **Intended use.** The grommets covered by this specification are intended for use in protecting the rotating band of separate loaded projectile during shipment and storage.

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 8140–A–146

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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Picatinny Arsenal
Dover, New Jersey 07801

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